THE SPACE RACE — From a Beeping Ball to a Home in Orbit (7.7, Part 1 of 2)

Student: Mark | **Duration:** ~60 min | **No textbook** — this lesson is built from videos, interactives, and discussion. The Spectrum text (7.7, "The World in Space") is the anchor of Part 2. | **Role: "Flight Director Mark"** — he runs Mission Control for this lesson, and gives the final ruling at the end.

The narrative spine: In 1957 the USSR launched a beeping metal ball the size of a beach ball, and the USA panicked. For 18 years two superpowers raced to humiliate each other — first satellite, first animal, first human, first woman, first spacewalk, the Moon — and every "first" was really a flex. But the race had a strange ending: after the shouting, the two enemies started building things together, and that cooperation is literally the house that Part 2 walks into.

The takeaway sentence Mark should be able to say at the end: "Space started as a race between enemies — and ended with them building a home together in orbit."

Resources

- [VIDEO] **Apollo 11 launch — crowd reaction version (HOOK CLIP, ~3 min, sound UP):** youtube.com/watch?v=2bztkoHmee0 (alternative, purely visual 4K high-speed: youtube.com/watch?v=cTYU6I8D9_w)
- [VIDEO] **TED-Ed, "Who won the space race?" (Jeff Steers, ~5 min, animated):** youtube.com/watch?v=FxpC-8f--xo (lesson page: ed.ted.com/lessons/what-was-the-point-of-the-space-race-jeff-steers; same talk at ted.com/talks/jeff_steers_who_won_the_space_race)
- [VIDEO] **Apollo–Soyuz Docking and Handshake | July 17, 1975 (~1.5 min, raw archival footage + original radio audio):** youtube.com/watch?v=LeA4dYZ_0no — used in Activity V; Leonov's famous line "Apollo and Soyuz are shaking hands now" is in English
- ☒☒ **Space Race Timeline** — custom HTML, `Space Race Timeline.html` in this folder (built, tested, sounds included)
- [HTML] **Newton's Cannon:** noragulfa.com/cannon (fallback: physics.weber.edu/schroeder/software/NewtonsCannon.html)
- ☒☒ Optional, for escape-velocity numbers: mechsimulator.com/tools/escape-velocity/
- ☒☒ Station pictures (used in **Part 2**, Activity VI — kept here so both plans are self-contained): **Salyut 1** (1971), **Skylab** (1973), **Mir** (1986), **ISS** (first module 1998) — NASA image galleries / Wikipedia
- [HTML] Bonus: **Kerbal Space Program** (paid game) — at-home suggestion; real orbital physics, his own rockets
- [VIDEO] Bonus: **Apollo 11 in Real Time** — apolloinrealtime.org/11/ — the whole mission replayable minute-by-minute

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ACTIVITY I — HOOK: THE BEEPING BALL (7 min) • video + guessing game

Video: "Apollo 11 launch — crowd reaction version" (~3 min) — youtube.com/watch?v=2bztkoHmee0. Sound UP, full screen. No intro, no vocabulary — play the clip first. Let it run. Let him react.

"Cool, right? July 1969 — the biggest rocket anyone had ever built, three men inside, going to the Moon. Here's your first question as Flight Director: how long before this do you think humans had ZERO things in space — not one satellite, not one rocket that had ever reached orbit? A hundred years? Fifty?" → (Expected: "fifty years?" / "a long time.") Reality: **twelve years.** In 1957 — closer to us today than 2014 is — nobody had ever put anything in orbit. Then a beach-ball-sized beeping ball changed everything.

"So how do you go from 'can't do it at all' to 'walking on the Moon' in twelve years? That's today's mission. The answer is: two countries got into the biggest fight in history — and neither one would quit."

1. Quick erudite showcase (2 min): "Before we start — show me what you've got. Name anything space-related you know." Let him run. Facts are currency; let him spend some. Park everything on a visible list — you'll tick items off as the lesson covers them.

Escape hatch: if he's flat, skip the showcase and go straight to the video — momentum matters more than completeness.

ACTIVITY II — THE RACE: TED-ED VIDEO (10 min) • video with checkpoint questions

Video: "Who won the space race?" — TED-Ed, Jeff Steers (~4.5 min, animated) — youtube.com/watch?v=FxpC-8f--xo (also on ed.ted.com).

Format: one video, paused 3 times. Rule before starting: "I'm pausing this. Your job is to catch facts — every time I pause, I need one fact you can give me." (Pause questions below are keyed to the actual transcript — the "wrong answers" are things that genuinely happen in the video.)

- **Pause 1** (~0:45, after the V-2 part): "Quick — the FIRST thing ever in space wasn't American or Soviet. Whose was it?" (Germany's — the V-2 rocket, a World War II missile.) "And what did the USA and USSR both do right after that war?" (Captured the rockets AND the scientists, and used them for their own projects.) Optional edge: "So the space race started with weapons?" (Yes — the R-7 that launched Sputnik was an intercontinental missile first. That's exactly why the beep scared everyone.)
- **Pause 2** (~2:00, after "Flopnik"): "America's answer to Sputnik — what happened on live TV?" (The Vanguard rocket rose two seconds and crashed.) "What did the newspapers call it?" ("Flopnik" / "Kaputnik"!) "And what did the Soviet delegate at the United Nations actually suggest?" (That the USA should receive foreign aid for developing nations — maximum mockery.) "Why were they laughing SO hard?" (They were ahead, and laughing is part of winning.)
- **Pause 3 / after video** (~3:30): "The video says the race lasted eighteen years — from stealing scientists to a handshake in orbit. Why do you think two enemies stopped racing and shook

hands?" (Expected: vague — money, tired, friends.) Hold it — Activity V answers it.

Comprehension check (verbal, quick):

1. What was the first human-made thing in space, and whose was it? (The V-2 rocket — Germany's, WWII)
2. What was actually scary about Sputnik — the ball, or something else? (The rocket: the same technology could deliver a nuclear warhead to any city)
3. What did newspapers call America's failed first satellite launch? ("Flopnik" / "Kaputnik")
4. Who got the first human into space, and who was the first American in ORBIT? (Gagarin, USSR, 1961 / John Glenn, USA, 1962)
5. Name two everyday things the video says came out of space technology. (Memory foam, freeze-dried food, LEDs used in cancer treatment, GPS/satellites)

ACTIVITY III — TIMELINE: BUILD THE RACE (12 min) • custom HTML

Format: Space Race Timeline (`Space Race Timeline.html`). Dark space theme, twinkling starfield (340 stars), sounds on. Three stages with a progress bar:

1. **Build the timeline — one year at a time.** The timeline computer shows **one glowing year at a time** on a horizontal strip at the bottom (it auto-scrolls as he works; ◀▶ arrows to peek). On screen: only **5 mission patches**, floating and centered at the top — each with a title, an emoji, and a **logic clue, no year**. Mark picks the patch that belongs at the glowing year: **tap a patch, then tap the year — or drag it down** (both fully iPad-safe). The 5 candidates are always **close together in time**, drawn from the years just ahead, so he has to actually read the clues — challenging but doable. Wrong picks cost one ☒☒ (3 per year) and grey out; after 3 wrongs, Mission Control points at the answer and he taps it to move on. **Placed patches never come back** — the pool shrinks until the last patch auto-places itself. Same-year events both fit the same slot. On success, the whole timeline appears as a clean scrollable review screen before step 2.
2. **Who did it?** — sort all 15 patches into USSR / USA / BOTH. Then the scoreboard appears: **USSR 7 (incl. 2 failed) • USA 6 (incl. 3 failed) • Both together 2.**
3. **The Verdict** — Mark votes who won the race (USSR or USA, big buttons), then the nuance reveal plays: either vote is defensible, the scoreboard is laid out, both sides lost astronauts, **the biggest victories came right after the biggest failures**, and the race ended in cooperation (Apollo–Soyuz → Mir → ISS). Final message: "Who really won? Science." Stars (1–3) by number of wrong tries; confetti + fanfares throughout.

The 15 events (for the teacher — the cards show title + clue, never the year):

Year	Event	Side	Failed?
1957	Sputnik 1 — the beeping ball	USSR	
1957	Laika, first dog in orbit	USSR	
1961	Gagarin — first human in space	USSR	
1961	First American in space (Shepard)	USA	
1962	"We choose the Moon!" speech	USA	

Year	Event	Side	Failed?
1965	First spacewalk (Leonov)	USSR	
1967	Apollo 1 fire	USA	☒
1969	Secret rocket explosion (N1)	USSR	☒
1969	Apollo 11 — first Moon landing	USA	
1970	Apollo 13	USA	☒
1971	Soyuz 11 tragedy	USSR	☒
1975	Handshake in orbit (Apollo–Soyuz)	BOTH	
1986	Challenger	USA	☒
1986	Mir space station	USSR	
1998	First piece of the ISS	BOTH	

Why it's built this way: only 5 patches on screen at a time (never all 15), and the 5 are drawn from nearby years — big enough field to make him think, small enough to stay winnable. The clues chain logically ("Once the beeping ball worked…", "Just WEEKS after the rival's flight…", "Tired of second place…"), so Mark can reason his way through even without background knowledge — reading clues out loud is part of the job. The 3-heart rule keeps stakes without dead ends: after 3 wrong picks Mission Control literally points at the answer, so the timeline always gets built. **Failures are deliberately on the timeline** (Apollo 1, the secret N1, Soyuz 11, Apollo 13, Challenger) — the road to space wasn't smooth, and the Verdict makes that a feature, not a footnote.

Framing line while it loads: "Mission Control, the timeline computer crashed. Fifteen patches, no dates. The clues on each patch tell you what came before what. Read them out loud — you're the Flight Director."

Teacher behavior: let HIM read every clue aloud and argue each placement. Don't confirm silently — his reasoning out loud is the activity. If he's stuck on one card, ask "What does the clue say happened right before this?" rather than giving the year.

Escape hatch: if the HTML misbehaves or he's itching to move, do the same 15 events as cards on the whiteboard/annotation tool — 90 seconds of dragging beats a broken activity. (Or drop to 8 core cards: 1957 Sputnik, 1961 Gagarin, 1967 Apollo 1, 1969 Apollo 11, 1970 Apollo 13, 1975 handshake, 1986 Mir, 1998 ISS.)

ACTIVITY IV — NEWTON'S CANNON: WHY IT'S HARD (11 min) • interactive simulation

Format: screen-share **noragulfa.com/cannon**. Frame it: "OK Flight Director, here's the problem the engineers had. Throw a ball — it falls. Shoot a cannonball — it falls. How do you make something NEVER fall?"

1. Let him shoot at low speed. "What happened? Why?" (It fell. Gravity.)
2. Crank the speed gradually. Watch it land farther and farther. "The ball is still falling the whole time. So what's changing?" (The ground curves away — Earth is round.)

3. Find the magic speed where it never lands (≈ 7.9 km/s). "It's still falling — it falls AROUND the Earth. That circle is called an orbit."
4. Turn on **"Show ISS"** in the sim: "That circle is where the space station lives — the one from next lesson. The ISS is falling right now, and so is everything in it. Remember that; we're coming back to it."
5. **Speed round (verbal math):** orbital speed ≈ 28,000 km/h ≈ **33 times the speed of sound**. "A fast car does 100 km/h. A plane about 900. The ISS does 28,000. How many times faster than a plane is that?" (≈30×.) "That's why rockets look like that — you're not climbing up, you're throwing something so hard sideways it keeps missing the ground."
6. (Optional if he's eating it up: mechsimulator escape-velocity tool — past ~11 km/s the rocket leaves Earth completely. That's Moon-shot speed.)

The chain to make him say out loud: fast enough sideways → falls around the Earth → orbit. Teaser: "So⋯ if the ISS is falling the whole time — why do astronauts float? Keep that question. It's next lesson's whole mystery."

Escape hatch: longest activity — if attention dips after step 3, skip steps 5–6. The orbit insight alone is worth the activity.

ACTIVITY V — THE VERDICT: WHO WON THE RACE? (8 min) • debate + the handshake footage

Video: "Apollo–Soyuz Docking and Handshake | July 17, 1975" (~1.5 min, raw archival footage with the original mission radio audio) — youtube.com/watch?v=LeA4dYZ_0no. Channel: Manned Space. There is almost no narration — it's the real radio traffic, so treat the audio as part of the activity, not background.

Format: back to the question from Activity II. Part 1 stays race-focused — the space-station lineup moved to Part 2, where stations are the whole point.

1. **The ruling first** (~3 min, no video yet): "The USSR won almost every 'first.' The USA won the Moon. Flight Director — your ruling: who won?" Let him argue it out with facts. (He already voted inside the HTML — here ask him to defend his vote, or play devil's advocate against it: "OK, but the other country would say⋯")
2. **Then the twist** (~2 min): "Whatever your ruling — here's how the race actually ended. Not with a winner. In 1975 these two ships met in orbit. Watch what the radio says." Play the docking/handshake video. Sound up.
3. **What he just heard** (~2 min) — the famous moment, all of it in ENGLISH: as the two ships locked together, Soyuz commander **Alexei Leonov** radioed: "Apollo and Soyuz are shaking hands now." When the hatches opened, the crews deliberately spoke **each other's languages** — Leonov greeted the Americans in English, and Apollo commander Tom Stafford answered in Russian with a thick Oklahoma accent. (Leonov later joked that Stafford spoke "Oklahomski." Kids love this detail.) If he asks what Leonov says: the greeting line he hears is English — the Russians saved their own language for talking to Moscow.
4. **The closer** (~1 min): "Eighteen years of racing — and it ended with a handshake. Why would enemies do that?" (Racing was ruinously expensive; together they could do more; maybe they were a bit less enemies by then.) One-line bridge: "And what they started building together

after that handshake — that's exactly where we go next lesson." (No station lineup here — that's Part 2's opening act.)

ACTIVITY VI — MISSION DEBRIEF (6 min) • verbal synthesis + bridge to Part 2

Format: the final log entry — no test energy, this is the mission log.

"Flight Director, final log entry. Finish this sentence: 'Space started as a race⋯'" → Goal: "⋯and ended with them building a home together in orbit." Help him compress his own version into that one sentence.

End on the weird-fact reward: the Saturn V from the hook clip was the most powerful rocket ever successfully flown — **for about 50 years**. It lost the crown to **SpaceX's Starship**: roughly TWICE the thrust, taller than the Saturn V, first launched in 2023, and its booster lands by being caught out of the air with the launch tower's giant mechanical arms. If Mark knows Starship or has watched a launch — this is his moment to flex. (Fact-check verified October 2026: Starship is the most powerful rocket ever flown, ~74 MN liftoff thrust vs Saturn V's ~35 MN.)

Bridge to Part 2: "Next lesson we stop racing and move in. That house they built together? We're going inside it — and we're going to find out why everyone inside it is floating."

Notes on Mark

- **He's the protagonist, named:** Flight Director Mark, explicit from minute one. Guide on the side — your lines are questions, not lectures.
- **Fast mode changes, nothing lingers:** clip → video → HTML → sim → discussion → debrief. If anything runs long, the escape hatches above are the plan, not a failure.
- **Verbal first, always:** every question is answered out loud before any clicking. He compresses ideas into one sentence with your help — that's a skill target.
- **The two anchor questions to protect:** "How do you make something never fall?" (Activity IV) and "Why do astronauts float if the station is falling?" (planted here, answered in Part 2 via PhET). If you keep nothing else, keep these.
- **Failures are content, not filler:** if Apollo 1 / Soyuz 11 / Challenger bring questions, answer them honestly and briefly — the HTML frames them as part of the road, and so should you. Tone: respect for the people, not shock.
- **Ride the tangent:** rockets, bombs, dogs in space, Moon-landing doubts — all welcome, 30–60 seconds, then "Good thought, Flight Director — log it and back to the mission."
- **China connection (optional, 30 seconds):** after the Apollo–Soyuz handshake, you can note that today China has its own space station (Tiangong) — and in Part 2 we'll check when it flies over his house.

Materials

- Screen-share ready: launch clip (I), TED-Ed video (II), `Space Race Timeline.html` (III), noragulfa.com/cannon (IV), handshake video (V)

- Whiteboard/annotation tool for quickfire lists and points
- No physical prep — the HTML is built and tested; just open it before class

Custom HTML needed

DONE — `Space Race Timeline.html` (this folder). Dark space theme, twinkling starfield (no drift — stars stay put and pulse), WebAudio sound effects (pickup/drop/chime/buzz/fanfares, toggle in header). **Step 1 is round-based like Ming's Medicine timeline: one glowing year at a time, 5 candidate patches close in time, pick the one that fits — tap-then-tap or drag, touch-friendly. 3 hearts per year, then Mission Control points at the answer; placed patches never return; the last patch auto-places.** Same-year tolerance, full-timeline review screen after step 1, side-sorting with live scoreboard, USSR/USA verdict vote, nuance reveal, confetti, star rating by total wrong picks. Fully rebuilt and playtested (desktop + iPad-size viewport, zero console errors). If you want anything tuned (pool size, hearts, clue wording, sounds off by default), say the word.